



Subcontracting in Unorganized Manufacturing: A Mode of Survival or a Strategy of Growth?

Samar Kumar Mishra^{1,2} · Bimal Kishore Sahoo¹

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Abstract

This paper investigates the incidence and determinants of subcontracting (SC) in unincorporated non-agricultural enterprises, the nature of the relationship and the net accumulation fund of firms undergoing subcontract. The incidences of subcontracting in India among small manufacturing firms have remained stagnant over time. However, there have been some changes within the categories. States with relatively poor infrastructure and industries that involve relatively low skill and technology have higher incidences of subcontracting. The profit share of SC firms is higher than those who do not subcontract, owing to low assets, location and ownership of firms. The higher profit share of the SC firms is because it consists of more OAEs. A considerable portion of the profit of OAEs comprises the wage of working proprietors and unpaid family workers. The net accumulation fund (NAF) of these firms is very low. Low NAF of SC firms is a major obstacle to firm size and output growth and is an important cause behind stagnant and contracting firms' decisions to subcontract. The study also finds that the subcontracting relationship and aftermath of own account firms (without hired workers) and establishments (with at least one hired worker) are different. The institutional intervention in terms of credit and provisions of subcontracting exchanges may lead the unorganized firms to grow and develop stable contracts for the benefit of firms and the economy.

Keywords Subcontracting · Manufacturing sector · Unorganized enterprises · Profit share · Net accumulation fund

JEL Classification D22 · L21 · O14 · O17

✉ Samar Kumar Mishra
mishrasamar89@gmail.com

Bimal Kishore Sahoo
bimalkishore.sahoo@gmail.com

¹ Department of Humanities and Social Sciences, IIT Kharagpur, Kharagpur 721302, India

² Department of Economics, Manipal University Jaipur, Jaipur, Rajasthan 303007, India

1 Introduction

Subcontracting links small and micro-enterprises with large industrial enterprises to the benefit of both. Subcontracting means the manufacture of goods by one firm for another based on the latter's specifications (Lazerson 1990a, b). A firm offering subcontract known as the principal firm requests another independent enterprise called the subcontracting firm to undertake the production or carry out the processing of a material, component, part or sub-assembly according to specifications or plans provided (Holmes 1986). It allows principal firms to take better advantage of the division of labour. Subcontracting helps principal firms shed marginal activities and concentrate on strategically important ones (Lazerson 1990a, b). The study explores subcontracting in the unorganized manufacturing sector in India. It examines the incidence and determinants of subcontracting and attempts to investigate the motivation of firms pursuing subcontracting.

Profit share is the simplest estimate of the firms' surplus and does not take into account the wages or compensation paid by the owner to family workers and other workers being engaged full time or part time. So, we estimate net accumulation fund (NAF) which provides a better picture of the firms' surplus. NAF is based on the notion that the owner possesses liability towards unpaid workers. The compensation to unpaid workers (pseudo wage) is calculated based on the wage of hired workers from similar enterprises. This pseudo wage is adjusted from the surplus of the firm in order to estimate the real surplus or NAF which potentially transforms to the capital accumulation by firms.

Subcontracting was not given due importance in early classical and Marxian literature, they considered it as a marginal activity which would gradually disappear. However, it continued across different phases of industrialization. Conceptualization of production techniques such as vertical integration and markets based production facilitates economic understanding of subcontracting (Arrow 1975; Coase 1937). A firms' decision to obtain a product either through internal organization or via the market depends upon transaction or organization cost. Firms rely on the market when the cost of organizing an extra transaction within the firm becomes equal to the cost of carrying out the same transaction by means of an exchange in the open market or cost of organizing in another firm (Coase 1937). Firms' ability to vertically integrate the production process depends on the ability to control and coordinate the high volume flows (Chandler 1977).

Subcontracting is not very widespread as it creates high transaction costs by the assets specificity linked to it. The asset's specificity arises in geographical, physical or human attributes. The transactions of these particular assets are highest, creating bilateral dependency. The problem of bounded rationality arising out of the dependency or bilateral monopoly involves additional settlement costs. The same is resolved very quickly in a vertically integrated company. Therefore, reliance on subcontracting is profitable only if the transaction costs are non-repetitive and the products require minimum capital or skill or if there are standardized and non-complex products (Williamson 1971).

In recent years, such linkages are steadily spreading in many industrially advanced regions. Small and micro-enterprises work for more extensive industrial units. The rationale for forging such an arrangement is robust only in some product lines. It is estimated that in 2015–16, 30.04% of manufacturing firms pursued subcontracting. Of the total, 32.64% of subcontracting firms were in rural areas against 26.51% in urban areas.

The unorganized sector enterprises employ families and other workers for production. These firms are either own account enterprises (OAEs) or establishments with few hired workers located within households. About 85% of the total unorganized firms are OAEs. There is a lack of information on the compensation paid to the family members. Therefore, there is a gap in profit or surplus reported and actual surplus. Surplus is a part of gross value added (GVA) left after paying rent, interest and compensation to the workers. The retained earnings or surplus of the enterprise is also the net income of the household (Kesar and Bhattacharya 2019). A part of the surplus is paid directly or indirectly to the family workers. This invisible part is a subset of the surplus of the enterprises. Involving family workers means more funds are required for consumption as the firms' income is the primary source of household earning. It is probably due to this factor that the reported surplus of the firms does not lead to the capital accumulation and enlarged productive capacity therein. The study, therefore, attempts to draw the net accumulation fund (NAF) using the pseudo wage fund calculated using a similar firm from the same cohort. Pseudo wage is the expected compensation given to the unpaid family workers, had they been working as a hired worker in the same enterprise.

The rest of the paper is organized as follows. Section 2 discusses the available literature on the issue, discussing both theoretical and empirical studies. Section 3, the methodology section, discusses the data, methods and variables used in carrying out required analyses. Section 4 consists of the results and discussion on incidence, profit share, pseudo wage and net accumulation fund, determinants of SC, the status of required assets and loans, and the nature of SC relationships. Section 5 discusses the conclusion of the study.

2 Review of Literature

There are many approaches to subcontracting in the literature. The early studies observe subcontracting in a dualistic approach. It is caused due to the labour differentiation based on skill and their positioning in the value chain system in heavy manufacturing. Large producers faced difficult challenges in manufacturing complexity and small firms were constrained by the technical and financial resources. To meet these challenges, large manufacturers began to invest heavily in their subcontractors (Nishiguchi and Brookfield 1997). In the flexibility approach, subcontracting reflects the search efforts on the part of large firms to gain margins flexibility in an increasingly competitive market (Ramaswamy 1999). In the transaction cost approach, the transactional environment consists of uncertainty, frequency in transactions and relation specific assets. A firm is to decide whether to internalize certain transactions or not? Subcontracting arrangements can be an

alternative to save the transaction cost. The network and the information approach are based on contract theory, particularly the principal agent model in which the long-term relationship works to ensure efficient risk sharing arrangements.

The debate around the rationale of subcontracting has two main perspectives, benign and the exploitation views. The benign view argues subcontracting as the alternative system to a vertically integrated system of production which leads to specialization and reduced production cost. It believes that the dynamic and modern segment of the small manufacturing firms carries out subcontracting (Basole et al. 2015; House 1984; Moreno-monroy and Erumban 2012; Uchikawa 2011). The exploitation view argues that the subcontracting reaps out all the excess labour of the traditional segment of the unorganized sector (Sahu 2010; Bhattacharya et al. 2013). In an unequal economic relationship, the chance of senior members behaving opportunistically is very obvious. It may take the advantage, either by breaking the contract or not including in the contracts the matters which violate their self-interest (Perrow 1986).

Subcontracting has been a model for industrial development in many developed countries. Small firms get the benefit of a network and information approach which helps in sharing their risk as well as brings stability (Taymaz and Kilicaslan 2000). The smaller firms which have succeeded with subcontracting have been able to turn out goods which meet the quality standard of their customer at a reasonable price. The firms that seem to be satisfied with the subcontracting are those having work of relatively lower precision or where they have had a long relationship with a few reliable subcontracting firms that know their customers' requirements and are willing to adjust them (Rosen 1956). Besides taking advantage of specialization and division of labour, the promotion of subcontracting is largely guided by two considerations: lower labour and overhead costs in small firms and restricting in-house employment of workers to minimize the potential threat from organized labour (Nagaraj 1984).

There is a significant positive relationship between formal sector subcontracting and total employment or number of enterprises in the relatively modern segment of the informal sector (Moreno-monroy and Erumban 2012). The Neo-Marxian literature of underdevelopment considers subcontracting as a system that uses capital to enhance the class exploitation of the reserve army which exists in the informal sector (Moser, 1978). The competitive capacity of subcontractors in managing their workforce, financial and managerial resources determines their success as in most cases, they are not provided with almost any kind of support in the modern form of subcontracting. The relationship is very unstable and ultimately beneficial to the large firms (Annabhujula 1988). Except for marketing assurance, small firms are not getting other forms of assistance such as technical, financial, supply of raw material and training. The performance of subcontracting exchanges has also not been satisfactory since large firms have their vendor development programmes (Sahu 2007). As we move from the principal firm to a small subcontractor firm, the composition of permanent labour decreases, and contract/casual labour increases. The nature and characteristics of labour relationships vary among the chains of subcontracting units and the worst case is of those engaged in the lowest end of the production chain (Annabhujula 1988). So there are arguments to enlarge the Indian Contract Act to

tackle the problem of moral hazards associated with subcontracting (Raychaudhuri 2004).

The large firms retain the core production while subcontracting the peripheral activities. The acquisition of specialist expertise, meeting fluctuations in demand along with the reduction in overhead costs and the ability to spread commercial risks are the commonly cited advantages of subcontracting in South Wales (Bresnen and Fowler 1994). Subcontracting is an indicator of the search efforts of firms to gain new margins of flexibility in competitive markets. It is expected that in due course of time there would be a qualitative shift in the nature of subcontracting perhaps with an emphasis on the use of skilled labour and specialization rather than on taking advantage of poor labour standards in the small-scale factories (Ramaswamy 1999).

The empirical study using data from NSSO survey of unorganized Manufacturing found the existence of subcontracting premium (Basole et al. 2014). Controlling for the difference in endowments, being in the subcontracting relationship is beneficial for the small informal manufacturing firms, though the effect is small. Furthermore, it is more beneficial for the relatively traditional or backward part of the informal sector than the modern part.

In food processing industry which observes high incidences of subcontracting, the workers employed in subcontracting Non-directory Manufacturing Enterprises (NDME¹) earned less than those of non-subcontracting. However, in Directory Manufacturing Enterprises (DME), the workers employed in subcontracting firms earned more than those employed in non-subcontracting (Ali 2021). The labour productivity in subcontracting firms was found to be less than those of non-subcontracting firms at aggregate level. Within subcontracting firms, the labour productivity of establishment firms is more than the OAEs (Singh 2020). The level of total factor productivity of subcontracting manufacturing enterprises also varies across the firm size, ethnic affiliation of entrepreneurs and subsectors (Hayashi 2002).

3 Empirical Framework

The study is designed to bring forth the incidence of subcontracting in Micro Small and Medium Enterprises along with selected indicators. It also aims to trace the possible determinants of subcontracting. The study further explores the nature of subcontracting relationships and retained earnings of firms undergoing subcontract.

3.1 Data

The study used data collected by National Sample Survey Office (NSSO), Ministry of Statistics and Programme Implementation (MoSPI) on ‘Unorganized Manufacturing Enterprises’ (62nd Round) and ‘Unincorporated Non-agricultural Enterprises’ (73rd Round) conducted in 2005–06 and 2015–16. NSSO 62nd Round

¹ As per NSSO, NDME is defined as a manufacturing enterprise employing less than 6 workers whereas an enterprise employing more than or equal to 6 workers is termed as DME.

data of Unorganized Manufacturing is based on the survey of 82.8 thousand firms conducted during July 2005 to June 2006 (National Sample Survey Office 2007). After incorporating list frame and area frame, data estimates 1.70 crores unorganized manufacturing firms in India across states and industry groups. NSSO 73rd Round data of Unincorporated Non-agricultural Enterprises is based on the survey of 2.90 lakhs firms conducted from July 2015 to June 2016 which consists of both semi-round 1 and semi-round 2. It estimates 6.43 Crores unincorporated firms out of which 1.96 crores (31%) are manufacturing enterprises (National Sample Survey Office 2018). Estimations are done using iweight.

The study uses 62nd and 73rd round of data as both of these round have same variable to capture subcontracting, i.e. 'did the enterprise undertake any work on contract basis? (yes-1, no-2)', {Item No. 229 in 62nd and 337 in 73rd round}. However, 67th round of NSSO does not have the same variable rather it has 'Whether the enterprise has any prior marketing agreement with other units? (yes—1, no—2)', {Item No. 233}. This makes it incomparable to the other rounds with regard to the subcontracting.

In the 73rd round, only industries classified as manufacturing at 2-digit in NIC 2008 have been considered, i.e. Section C and Division 10 to 33. The values of assets and income have been converted to the 2004–05 base year using WPI values provided by the RBI. For consistency of the two industry classifications, the 2-digit concordance provided by the KLEMS (RBI) has been followed in the study. The study further uses Gross State Domestic Product (GSDP) per capita values drawn from RBI handbook of statistics and the district level enterprises values from 6th economic census (2013–14).

3.2 Methods

Incidence or the prevalence of subcontracting has been calculated using descriptive statistics and growth rates. The determinants of subcontracting, the study uses binary outcome Model for estimation of pooled series of NSSO 62nd (2005–06) and 73rd round (2015–16) data. The model uses Logit regression as it has low AIC and BIC values. Four models have been developed using the combinations of dummies for the Time, State and Industry. Marginal effects are used to draw interpretation. Subcontracting (SC) decision, a binary outcome variable is a dependent variable in the Model. The dummies for location of enterprise, type of enterprises, sector, status of growth, ownership of enterprises, whether pursuing mixed activity, status of registration along with the log of net accumulation fund and firms' total assets have been the dependent variable in all the models. Dummies of States and Industries have been used as per the Model. To control for the extreme values, only firms with NAF in the range of -5 lakhs to +50 lakhs are taken into consideration into the model. It drops 0.35% of the total observation from the model and makes data more representative and meaningful. Further, the study also develops a binary outcome model on the likelihood of a firm being subcontracted using the CAGR of GSDP per capita along with the district's share of unorganized and organized manufacturing firms as independent variables.

NAF is calculated using the following equation:

$$\text{Pseudo wage} = \text{Pseudo wage of OAE} + \text{Pseudo wage of Establishment} \quad (1)$$

$$\text{Total wage} = \text{wage (working owner + hired worker)} + \text{Pseudo wage of other workers} \quad (2)$$

$$\text{NAF} = \text{GVA of Firm} - \{\text{Total wage} + \text{Interest paid} + \text{Rent paid}\} \quad (3)$$

The study imputes pseudo wage independently of OAEs and establishment. In establishments (at least 1 hired worker), the pseudo wage of other workers or workers is the average wage of the hired worker in the same enterprise. This pseudo wage is multiplied by the number of other workers to draw the pseudo wage fund of the establishments. For own account enterprises (with no hired worker), pseudo wage is estimated from the similar enterprises in the establishment category. A similar enterprise is an enterprise with similar fundamental characteristics. For this purpose, we divide the entire group of firms into various cohorts. Cohorts have been developed considering the industry group (2-digit), sector (rural/urban), the type of firm (OAE/establishments), number of hired workers and GVA per worker.

The entire data point of each round is first divided into a group of firms of the same industry group within the same sector and having similar GVA. The present study is based on 14 industry groups as per KLEMS;² therefore, we have 28 broad categories (14 for each rural and urban). Similar firms require firms in the same cohort with GVA per worker in the $\pm 5\%$ in the same category. For the pseudo wage of OAEs of a particular cohort, we draw the average wage of hired workers of the establishment firm in the same cohort having only 1 hired worker and GVA per worker of $\pm 5\%$. In cases where there is no establishment firm in a cohort against an OAE having 1 hired worker and GVA in $\pm 5\%$ range, then we consider firms with 2 hired workers in the same cohort.

3.3 Variables

The probability of a firm being in subcontract is given by:

$$\begin{aligned} P_i &= E(Y = 1 | X_1, X_2, \dots, X_k) = F(\beta_0 X_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k) \\ &= \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}} \end{aligned}$$

where

$$F(x) = \frac{1}{1 + e^{-x}}$$

² Reserve Bank India research projects on Capital (K), Labour (L), Energy (E), Materials (M) and Services (S) inputs.

Table 1 List of variables incorporated in the model

Label	Description	Expected sign	Explanation
Location	Within household = 0 @Outside household = 1	–	Small units are low in endowment
Type	OAE = 0; @Establishment = 1	–	Low resources, low production capacity
Sector	Rural = 0; @Urban = 1	–	Higher concentration of large firms in urban areas
Status	Contracting = 0; @ (Expanding + Stagnant + operated less than 3 years) = 1	–	Firms are expected to gain & expand from SC
Ownership	Female owned = 0; @ (Male owned + owned by others) = 1	+	Benefit linked with marketing agreement
Mixed activity	@Not pursuing mixed activity = 0 Pursuing Mixed Activity = 1	+	Reflects dynamic entrepreneurship skills
Registration of enterprises	@Not registered = 0; Registered = 1	+	Registration induces forward linkages
Net accumulation fund	Log of (real) funds accumulated by the enterprises	+	Linked to the growth of the firm
Total assets	Log of (real) assets acquired by the enterprises	+	SC is linked to the economies of scale
State dummies	Yes = 1		
Industry dummies	Yes = 1		
NSDP per capita	CAGR of NSDP from 2004–05 to 2013–14	+	High growth of states is induced by higher industrialization
Share of unorganized sector enterprises	District wise share of unorganized enterprise in total enterprise	–	Clusters have no role in determining SC
Share of organized sector enterprises	District wise share of organized enterprise in total enterprise	+	Large and small firm linkage

Table 2 Incidence of subcontracting across selected dimensions In percent (%) *Source:* Authors' calculation based on NSSO (62nd & 73rd) rounds; *Note:* Growth depicts the change in total number of subcontracting firms

	Sector		Type of ownership		Entp. type		Location of entp	
	Rural	Urban	Male owned	Female owned	OAE	Estb	Within HH	Outside HH
2005–06	30.4	34.7	20.5	50.4	32.5	26.8	37.4	16.3
2015–16	33.3	27.5	19.4	44.8	32.4	21.7	37.0	15.1
Growth	3.7	33.8	−3.7	25.9	15.7	−4.7	13.4	12.4

The binary-dependent variable Y (subcontracting status) is determined by the set of explanatory variable shown in (Table 1).

4 Results and Discussion

4.1 Incidence of Subcontracting

Before going into the details of the subcontracting model prevailing in India, it is important to analyse and explore the incidence of subcontracting. Incidence is the simple prevalence across selected dimensions. In this section, we investigate the prevalence across sector, type of ownership, type of enterprise and location of enterprises as shown in Table 2.

There has been an absolute growth in the number of SC firms in the urban areas. However, the incidence has declined. The growth in non-subcontracting (NSC) firms is much higher than the growth of SC firms. For the rural firms, it is the other way round. Firms located in rural areas also face credit shortages due to a low concentration of financial institutions. Firms prefer to switch or establish in urban or semi-urban areas due to the availability of resources and markets.

The high growth of SC firms in the urban areas is the result of its linkage with the growth of formal and informal sector growth which will be discussed in the later section. The rise in incidence across rural areas can be accredited to the expanding rural market along with rising entrepreneurship (Sahu 2010). Further, the male-owned firms, especially establishments, have witnessed a decline in both the number and incidence of firms. OAEs and female-owned enterprises are more into SC. Firms located within the household have more incidences of SC as female-owned own account enterprises of the rural sector are mostly located within households. The rural firms, own account firms and the one owned by females are mostly of low assets and also lack other resources to carry out the full cycle of purchasing raw material, production and marketing. SC is therefore expanding in a very limited space and is confined mainly to the vulnerable firms, i.e. rural own account firms owned by females and located within the household.

Table 3 Status of manufacturing and subcontracting firms. In Percent (%) *Source:* Authors' calculation based on NSSO (62nd & 73rd) rounds; Note1: Gr. Rate is the compound annual growth rate of the number of firms; Note 2: Firms operated less than 3 years have been termed as new firms

Status of firm	Manufacturing firms			Subcontracting firms		
	2005/06	2015/16	Gr. Rate	2005/06	2015/16	Gr. Rate
Expanding	19.06	26.05	50.41	21.43	20.48	3.24
Stagnant	53.67	48.69	− 0.15	48.99	53.5	17.96
Contracting	15.30	14.5	4.38	14.31	17.35	30.98
New firms	11.98	10.76	− 1.11	15.28	8.67	− 38.72

In the survey of the unorganized sector, firm's status of growth is categorized as expanding, stagnant, contracting and others (operated < 3 years) based on the performance during the last three years (Table 3).

The share of expanding firms has increased in the manufacturing sector significantly. However, its share has decreased within subcontracting firms. The growth in the number of expanding firms in subcontracting is also very marginal. Further, the growth in the number of stagnant and contracting firms is very high among subcontracting firms in comparison with the overall manufacturing firms. Thus, the major fraction of the subcontracting firms is either stagnant or contracting. This proportion is further likely to get intense given their higher growth rates. It reflects that the subcontracting mode of production does not lead the firms to take advantage of the production scale. Firms do not witness any incentive for working under the subcontract. It is possibly pursued as a model of the scarce resource firms. The later section of the study has also discussed the nature and type of subcontracting prevailing in India and the contracts that the principal firm and the subcontracting firms share.

Firms' production and marketing strategy is governed by the state's policy and business environment. The size of the formal and informal sector along with the infrastructure, communication and resources in a state plays an important role determining SC status. Incidences across the states have been depicted (Table 4) to analyse the regional variances.

The incidence is strong in many states, but only a few states have shown a rise in subcontracting incidences. States which are dominant in manufacturing like Maharashtra, Tamil Nadu and Gujarat have declining incidences. This may be because these states have relatively better infrastructural facilities like transport and warehousing which helps firms to carry out all their production and related work. It is important to note here that Maharashtra (21.46%), Gujarat (14.60%) together constitutes more than one third of the country's share of manufacturing sector GVA (Annual Survey of Industries 2013–14, 2016).³ Subcontracting therefore is not popular among states on high growth terminus.

³ Annual Survey of Industries (ASI) is carried out every year by the National Statistics Office (NSO) and it covers factories registered under Sect. 2(m)(i) and 2(m)(ii) of the factories act 1948.

Table 4 Incidence of SC across major states. In Percent (%)
Source: Authors' calculation based on NSSO (62nd & 73rd rounds; Note: States with less than 100 subcontracting firms are eliminated for comprehensibility; data for the state Telangana in 2015 have been merged with Andhra Pradesh

State	2005–06			2015–16		
	Rural	Urban	Total	Rural	Urban	Total
Jammu & Kashmir	29.2	10.2	25.6	13.1	5.3	10.2
Himachal Pradesh	3.1	7.1	3.3	0.7	7.3	1.5
Punjab	18.0	27.9	22.8	1.7	15.2	9.0
Uttaranchal	4.7	4.5	4.6	13.3	3.5	9.6
Haryana	3.0	11.5	7.1	3.4	8.6	6.1
Delhi	52.6	51.7	51.7	8.6	36.8	36.6
Rajasthan	8.0	24.7	14.2	1.0	1.8	1.5
Uttar Pradesh	35.6	34.6	35.3	15.6	34.7	23.3
Bihar	17.4	19.8	17.8	6.7	24.4	11.9
Manipur	31.5	22.7	28.7	3.0	2.2	2.7
Tripura	18.5	8.3	17.0	13.3	27.7	17.8
Assam	11.6	17.8	12.2	12.0	18.3	13.2
West Bengal	51.8	65.1	54.3	65.7	62.6	64.9
Jharkhand	48.8	21.3	46.7	34.5	4.0	30.8
Orissa	17.5	14.0	17.2	25.0	18.9	24.0
Chhattisgarh	3.5	12.8	5.0	3.7	9.7	5.6
Madhya Pradesh	3.9	21.4	9.8	25.9	26.5	26.2
Gujarat	11.4	35.4	24.4	5.2	19.2	15.9
Maharashtra	13.7	26.0	19.9	5.1	24.7	15.4
Andhra Pradesh	20.0	28.6	22.5	39.2	20.4	31.4
Karnataka	42.1	31.9	38.9	35.9	13.9	26.4
Kerala	24.5	33.5	26.8	20.1	21.4	20.8
Tamil Nadu	55.5	48.1	52.4	34.6	30.9	32.2
Total	30.4	34.7	31.7	33.3	27.5	30.9

The Micro Small and Medium Enterprise (MSME) of West Bengal is dominated mainly by the Tobacco, Textiles and Leather industry.⁴ These industries are the ones that witness the highest incidences of SC (Table 5).

Only selected industry groups have shown a rising incidence of SC. Low technology and low skill-oriented industries are more into subcontracting. Among industries, basic metals (13.24%), coke and refined petroleum products (10.75%) and chemical and chemical products (8.75%) have high share of GVA, whereas food products (11.65%), textiles (11.05%) and wearing apparel (7.23%) has highest share of employment (Annual Survey of Industries, 2013–14). It is therefore evident (Table 5) that the subcontracting is linked more to the labour-oriented industries. Industries requiring relatively high technology, skill and the one with multiple production chains have either low or declining incidences of subcontracting. The result is contrary to the experiences of successful SC models in developed economies like

⁴ 28 Percent of all manufacturing MSMEs in the Food, Beverage, and Tobacco industry and 42.4 percent in the Textile and Leather industry is located in West Bengal.

Table 5 Incidence of SC across major industry groups. In Percent (%) *Source:* Authors' calculation based on NSSO (62nd & 73rd) rounds

Industry group	2005–06			2015–16		
	Rural	Urban	Total	Rural	Urban	Total
Food & beverages and tobacco	39.6	30.6	37.9	54.2	38.3	49.9
Textiles, textile products and leather & footwear	58.4	60.1	59.0	61.5	58.6	60.0
Wood and of wood and cork	8.5	15.2	9.2	3.9	10.5	5.3
Pulp, paper & paper products and printing & publishing	73.0	39.0	53.5	27.5	31.3	30.5
Chemicals and chemical products	74.6	77.9	75.6	46.0	23.0	37.0
Rubber and plastics	31.4	36.6	34.8	70.1	40.0	51.3
Other non-metallic mineral	6.1	11.9	7.1	23.8	26.6	24.5
Basic metals and fabricated	15.2	37.6	24.5	5.3	20.2	14.6
Metal products						
Machinery, n.e.c	11.5	35.6	26.1	2.3	25.6	17.8
Electrical and optical equipment	12.2	18.2	15.5	1.2	4.9	3.5
Transport equipment	45.0	25.3	31.2	16.8	31.1	29.1
Manufacturing n.e.c., Recycling	23.1	35.7	29.9	24.8	26.0	25.4
Custom tailoring	6.0	13.3	8.7	2.4	4.6	3.4

Japan and Korea. This also provides a glimpse of the model which is not linked to the specialization but is driven by the availability of low-cost labour in the unorganized sector. With respect to either stagnant or declining incidence of subcontracting across high technology and skill-oriented industries, it can be partly manifested that the subcontracting in India has not led to specialization or have not become part of global value chains. With rising globalization, the state of technology and requirement of skilled man power will only increase. India aspiring to become global leader in manufacturing needs to promote linkages in a way to accommodate small and marginal producers owing to the advantages in terms of flexibility and cost advantages along with employment creation at local levels. It is further, important to figure out the profit share and net accumulated fund of the firms to analyse the absolute benefit from SC.

4.2 Profit Share of Firms

The profitability of the firm is important for its survival and also crucial for the livelihood of the firm owner (Das and Das 2017). The role of MSMEs in the development of societies relies on the profitability of the firms. NSSO does not provide data regarding profit share directly; however, it can be drawn from the net surplus. Net surplus is calculated by subtracting the factor income like wage, rent, profit and cost of raw material (Das and Das 2017). Profit share⁵ is the ratio of net surplus to the respective GVA. The

⁵ Profit Share = (Net Surplus/GVA)*100.

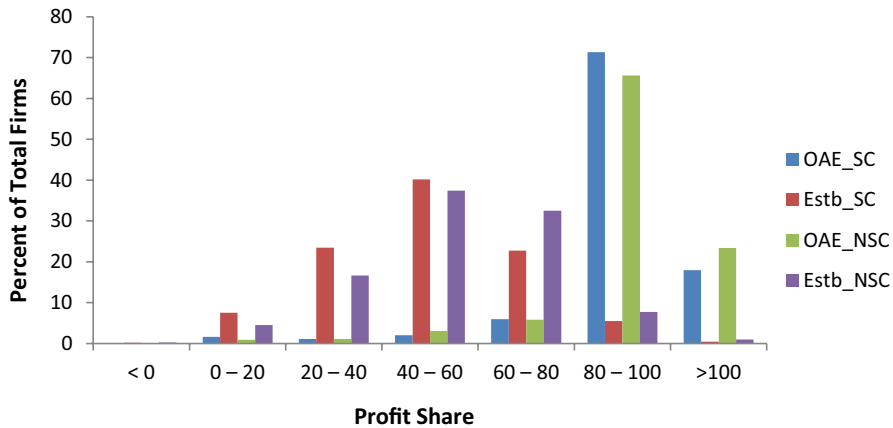


Fig. 1 Profit share across type of enterprises. (in %) *Source:* Authors calculation using data from NSSO 73rd round

Table 6 Pseudo wage—monthly (median value). (Rs. in whole number) *Source:* Authors’ calculation based on NSSO (62nd & 73rd) rounds

Type of firm	Subcontracting firms		Non-subcontracting firms	
	2005–06	2015–16	2005–06	2015–16
Establishment	1600	3910.9	1500	3259.1
OAEs	1261.94	2865.5	1471.4	2918.8

data of GVA and net surplus is collected by the NSSO with a recall period of 30 days, it can, however, be multiplied by the ‘Number of Months operated’ to get the annual values. The annual profit share of firms across type of enterprises is exhibited (Fig. 1).

The OAEs have a higher profit share in comparison with the establishments for both SC and NSC firms. There exists a significant difference in the profit share of OAEs and establishments. Among OAEs, SC firms report relatively higher profit share. The type of enterprise plays a larger role in determining profit share than the status of subcontracting. Lower assets and GVA of OAEs are the reasons behind high-profit shares. Also, OAEs mostly operate from home and have no hired workers. The net surplus of the OAEs also includes the wages of unpaid family workers and working owners (Kesar and Bhattacharya 2019). The owner of the firm is, however, obliged to compensate the family workers directly or indirectly. It is therefore more realistic to measure surplus value by controlling or adjusting the compensation of the family workers.

4.3 Pseudo Wage (PW) and Net Accumulation Fund (NAF)

NAF is a relatively more realistic reflection of firms’ surplus over a period. The unorganized sector is largely dominated by firms with either no hired worker (OAEs) or with only 1 hired worker. These firms are mostly home-based with the involvement

Table 7 Real annual NAF (median value). (Rs. in whole number) *Source:* Authors' calculation based on NSSO (62nd & 73rd) rounds

Type of firm	Subcontracting firms		Non-subcontracting firms	
	2005–06	2015–16	2005–06	2015–16
Establishment	36,564	128,356.3	35,340	103,508.9
OAEs	–6471.5	11,419.9	–2774.3	25,655.6

of family members who are unrecorded and unpaid. In traditional home-based enterprises, the income of the enterprise is mostly the main source of livelihood for the whole family. The surplus of the firms therefore could not get translated into the firm's capital accumulation. The study, therefore, estimates pseudo wage which the enterprise is expected to pay the family workers had they been hired systematically. It is the reflection of firms' liability towards unpaid family workers as reported in Table 6.

However, the owner of the enterprise might have paid a lesser wage to the family worker in comparison with the hired worker. Hired workers work for full 8–9 hours a day and are relatively more efficient and/or skilled. Pseudo wage calculated based on the wage of hired workers might be overestimated or underestimated in some cases. The NAF calculated accordingly is presented in Table 7.

For OAEs, both subcontracting and non-subcontracting, the value of NAF turns out to be negative. However, the values of subcontracting OAEs are much lower than that of non-subcontracting OAEs. It is contrary to the status of profit share. We have seen in the preceding sections that the relatively low assets and vulnerable category firms are more into the subcontracting mode of production. Most of the firms located within households generated value only sufficient to fulfil daily subsistence. The family workers are also underpaid and there is almost no scope for the growth of the firms.

On the contrary, the NAF of subcontracting establishments is more than those of the non-subcontracting ones. Establishments are located mainly in urban areas and have operational advantages. They are better positioned to take advantage of the subcontracting system and other institutions. In the later section, the status of loans and assets acquired has been discussed to clarify the situation. Also, NAF is a crucial output factor that may act as a deciding factor behind the firms' behaviour. The study incorporates NAF as a critical variable in analysing the determinants of subcontracting. NAF is an outcome variable and reflects the performance of the enterprise and the way forwards.

4.4 Determinant of Subcontracting

The preceding section dealt with raw averages regarding the incidence of subcontracting on various grounds. The following section discusses the determinant of subcontracting. The primary interest in doing the same is to bring the impact of some selected variables on the likelihood of a firm being in subcontract (a binary response, 'Yes' or 'No'). Firms' decision to subcontract depends on several factors.

Empirical studies have found a higher incidence of subcontracting in the relatively traditional and poorly endowed firms of the relatively less industrialized states in the manufacturing sector (Basole et al. 2015; Sahu 2010). Home-based, female-owned and asset-poor firms are more likely to subcontract than their counterparts. Engaging in the subcontracting relationship under the circumstances of limited working capital, product designs and markets benefits a group of small firms with ready access to working capital as well as markets which enables fuller capacity utilization. The findings are against the benign view of subcontracting. The descriptive statistics discussed in this study also highlight the consistent results using recent data.

In India, only a section of unorganized firms registers themselves under various acts and authorities to ensure the prospective benefits. Registration of firms is one of the major components of backward linkage and it enables to develop the forward linkage with the formal sector (Arimah 2001). A registered firm is expected to be a part of the subcontracting system as per the studies. Most of the benefits of being registered are concentrated to provide a suitable market and other benefits relating to working capital and raw materials. A registered firm also finds it easy to get a loan from the organized financial institutions. In a similar line, a firm's ability to pursue mixed activity⁶ reflects the dynamic entrepreneurial skills of the owner.

The empirical studies, as well as the descriptive study on subcontracting, have results against the theoretical argument. The dynamic and modern segment of manufacturing is less into subcontracting in comparison with its counterparts.

Further, the study designs a confirmatory analytical model using the concerned variables. A binary outcome model has been developed to evaluate the role of various factors in determining subcontracting incidences. The model is based on the pooled data of 2005–06 and 2015–16 provided by the NSSO. The model uses dummies for location, type, sector, ownership, the status of firms and status of firm's registration and firm's decision to carry out mixed activity along with the log value of real assets. The status of the firm is categorized as contracting and non-contracting. The study includes the firm's decision to pursue mixed activity and registration, as these are the crucial indicators of the dynamic segment of the industry. The model takes state and industry dummies to control for the fluctuations by the same. The logit model developed accordingly has been estimated through maximum likelihood estimation procedures. The marginal effects and standard error (robust) of the variables taken into the model have been depicted in Table 8.

Over time subcontracting is declining in the manufacturing sector, although the absolute numbers seem to rise over time. Female-owned firms located within the household are more likely to subcontract than male owned. Women do have to engage themselves as homemakers in most households. Also, in rural areas, transport and communications are a major hindrance to women's participation in marketing and allied activities. Against this, rural and own account enterprises are less likely to subcontract than their counterparts. A firm with declining output, i.e.

⁶ As per NSSO, it is an act of carrying out multiple activities. In the surveys of NSSO, enterprises are considered to have mixed activities if their activities are different at 2- digit respective NICs.

Table 8 Marginal effects of the selected variable. *Source:* Authors' calculation based on NSSO (62nd & 73rd) rounds

	Model 1 (Without T and State & Industry Dummies)	Model 2 (Without T and with State & Industry Dummies)	Model 3 (With T and Without State & Industry Dummies)	Model 4 (With T and State & Industry Dummies)
Rural	-0.077*** (0.0146)	-0.0501*** (0.0163)	-0.0715*** (0.0147)	-0.0457*** (0.0164)
OAE	-0.1178*** (0.0190)	-0.0619*** (0.0205)	-0.1131997*** (0.0192)	-0.0579*** (0.0207)
Within household	0.0861*** (0.0174)	0.0513*** (0.0187)	0.0883326*** (0.0176)	0.0535*** (0.0188)
Female owned	0.0515*** (0.0171)	0.0770*** (0.0221)	0.0556279*** (0.0171)	0.0799*** (0.0221)
Contracting	0.0250*** (0.0193)	0.0116*** (0.0213329)	0.0220702*** (0.0195)	0.0102*** (0.0214)
Pursuing mixed activity	-0.0784*** (0.0431)	-0.0663*** (0.0458)	-0.0800544*** (0.0430)	-0.0698*** (0.0458)
Registered	0.0103*** (0.0196)	-0.0039 (0.0217)	0.0086694*** (0.0198)	-0.0051** (0.0218)
Log of total assets	-0.0259*** (0.0051)	-0.0163*** (0.0060)	-0.019031*** (0.0057)	-0.0108*** (0.0065)
Log of net accumulation fund (Annual)	0.0124*** (0.0420)	-0.0046 (0.0446)	0.0197*** (0.0388)	0.00269 (0.0425)
Time variant (T)	-	-	-0.0472*** (0.0160)	-0.0391*** (0.0182)
State dummies	N	Y	N	Y
Industry dummies	N	Y	N	Y

Values in parentheses represents robust standard errors; *** Values significant at 1% ($p < 0.01$), ** Values significant at 5% ($p < 0.05$), * Values significant at 10% ($p < 0.10$)

Table 9 Marginal effects and the robust standard errors. *Source:* Authors' calculation based on data from NSSO 73rd round, RBI handbook of statistics and 6th Economic Census

Variables	Model 1	Model 2
Net state domestic product per capita (CAGR of 2004–05 to 2013–14)	– 0.0045*** (0.0063)	– 0.0049*** (0.0062)
Share of unorganized manufacturing enterprises (2013–14)	0.0007*** (.0008)	–
Share of organized manufacturing enterprises (2013–14)	–	0.0011*** (0.0006)

Values in parentheses represents robust standard errors; ***Values significant at 1% ($p < 0.01$), **Values significant at 5% ($p < 0.05$), *Values significant at 10% ($p < 0.10$)

contracting firms are more likely to subcontract than expanding and stagnant. A firm's involvement in mixed activity decreases its chances of being subcontracted. A registered firm is likely to subcontract but only in a limited industry and state. While controlling for state and industry fluctuation, registration status decreases the likelihood of subcontracting. Registration of firms is more evident in the advanced states like Maharashtra, Gujarat, Karnataka, Kerala and Andhra Pradesh owing to the awareness and relatively better public institutions. Further, as evident in the model, the likelihood of being subcontracted decreases as the asset of the firm increases.

Controlling for the state and industry, the NAF do not have any significant role in determining the firms' decision to subcontract. Therefore, the endowment factors like the type of firm, ownership, location, sector and assets do play a key role in determining firms' choice to subcontract. On the contrary, the expected surplus does not play any significant role. It is further important to analyse whether SC is linked to the states' growth? The study also attempts to analyse here the role of competition and of formal enterprises locally. An unorganized firm's decision to subcontract may be induced by the competition at the local level. The resource-scarce firms find it difficult to carry out both marketing and production work by themselves or it may also be a part of their strategy to have a marketing agreement and specialize in production activity. On the other hand, the large firms also induce the smaller firms to work for them. It establishes formal informal linkages through ancillarization or clustering. Accordingly, the study develops the logit model using growth in NSDP per capita along with number of unorganized and organized firms alternatively in two different models. The share of enterprises here is the percentage of total (respective state) enterprises located in the particular district.

With growth in NSDP per capita, the likelihood of a unorganized firm being in subcontracting declines (Table 9). This is consistent with the previous results of this study and further weakens the growth and benign approach to subcontracting. Further, the presence of both the unorganized and organized enterprises promotes subcontracting; however, the presence of organized manufacturing enterprises has a more vigorous impact (higher marginal effect). It therefore establishes the theory of organized-unorganized linkage theory of the subcontracting. However, the linkage is valid only in the states with low GSDP per capita or less industrialized states.

Table 10 Real net addition to owned assets during reference year. (Rs. in whole number)
Source: Authors' calculation based on NSSO (62nd & 73rd rounds)

	Subcontracting firms		Non-subcontracting firms	
	2005–06	2015–16	2005–06	2015–16
Establishment	33,697.2	133,191.1	35,652.7	17,555.4
OAEs	932.4	684.4	2172	2603.7

Sources of Loan

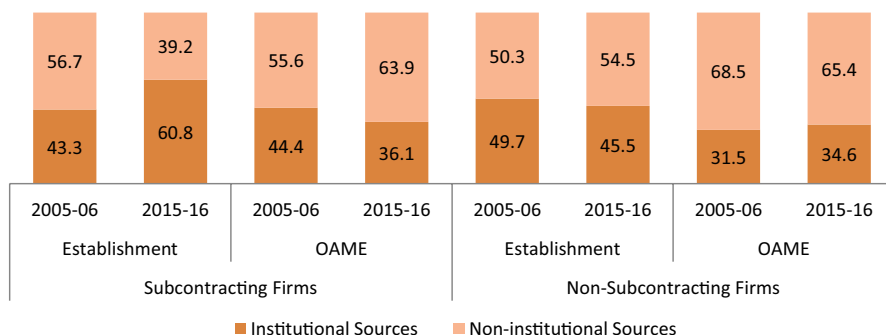


Fig. 2 Loan sources of the enterprises. (in %) *Source:* Authors' calculation based on NSSO (62nd & 73rd rounds)

4.5 Status of Assets Acquired and Loan

Given the status quo of prevailing subcontracting in India, it is important to further analyse whether and to what extent the enterprises are able to accumulate assets or capital. The acquisition of capital is an important reflection of economic growth and sustainable industrialization.

There is a significant gap between the assets acquired by OAMEs and establishments. It is worth noting that while subcontracting establishments have added the highest among the groups, the subcontracting OAMEs have added very nominal and even less than the previous year (Table 10). As discussed before, low assets enterprises prefer to subcontract, it is clear here that the own account subcontracting enterprises have not been able to acquire assets over the years. Having negative NAF, own account firms are expected to have lower assets acquisition over years.

It has been observed that the MSMEs do have very limited access to finance, especially institutional finance. Around 15% of the firms reach out to take a loan and most of them from non-institutional sources. It is therefore often observed that the loan amount is underreported by firms, especially those taken from non-institutional sources. For the sake of simplicity, all the sources of loans are categorized in this study as institutional and non-institutional. The institutional consists of central and state level term lending institutions, government (central, state, local bodies), commercial banks, cooperative banks and societies, micro-finance institutions and other

Table 11 Type of contract. (in %) *Source:* Authors calculation using data from NSSO 73rd round

Type of contract	OAE	Estb	Total
Solely for contractors	88.92	66.91	86.67
Mainly for contractors	6.16	17.09	7.28
Mainly for customers	4.92	16.00	6.05
Total	100	100	100

Table 12 Sources of equipment & raw material.(in %) *Source:* Authors calculation using data from NSSO 73rd round

Sources	Supply of equipments		Supply of raw materials		Design specifications	
	OAE	Estb	OAE	Estb	OAE	Estb
Self-procured	75.65	89.21	4.76	13.47	10.78	11.95
Supplied by the contractors	14.59	7.79	89.13	71.12	89.22	88.05
Both	9.76	3.00	6.11	15.41	–	–

institutional agencies. Non-institutional sources consist of money lenders, business partner(s), suppliers/contractors, friends and relatives, and others (Fig. 2).

We find a similar trend of difference between the behaviour of OAEs and establishments. OAEs rely more on non-institutional sources. It is also because most of the OAEs are not registered and have very low assets. They fear the formal process done at the banks and cooperatives. The loans taken from the non-institutional sources are mostly of very small amounts and are not aimed at acquiring capital for the enterprises. Own account firms, especially subcontracting ones, are therefore trapped in a low-level equilibrium trap. Low output and capital nexus can only break with investment and subsequent growth in the capital formation.

4.6 Nature of Subcontracting Relationship

In the above sections, we have explicitly discussed the nature of firms that pursue subcontracting along with the factors that likely influence the firms' decision to subcontract. The role of subcontracting in firms' growth along with the industrialisation of the economy at large also depends on the type of contract that firms undergo. The study draws the type of contract in the year 2015–16 (Table 11).

The nature of the contract varies along with the type of enterprise. A relatively large number of enterprises in establishments prefer to market their products beyond subcontracting agreements. OAEs have relatively lower output owing to low human and working capital. Therefore, they are mostly intact to the clients under a prior contract. Furthermore, firms' dependence on equipment, raw material and design is also an indicator of the sustainable subcontracting relationship. The biggest advantage of the system is that it may lead to a transfer of technology, skill and will therefore lead to the industrialization of the economy (Table 12).

Raw materials are the main assistance that subcontracting firms receive from the principal firms. The equipment with the SC firms is mostly self-procured. However, the own account enterprises (OAEs) receive greater assistance of both raw material and equipment in comparison with the establishment firms. OAEs are mostly low assets firms and are therefore dependent on either principle firms or the government to carry out their production. Establishments market their products beyond subcontracting agreements also and therefore prefer less to make agreements regarding raw material and equipment. Almost all SC firms receive design specifications from the principal firms. There is no difference in design specification along with the type of firms. For a large firm, subcontracting is a part of their vertical production. The design specification is, therefore, an integral part of the contract. Thus along with marketing assurance, firms receive raw material and design specifications from the principal firm. The supply of equipment is very limited and so are technological linkages. While this gives a limited advantage to the small firms under a subcontract, it may not lead to the growth of firms and the industrialization of the economy at large. A sustainable model requires a close and cordial relationship which will benefit both in the long term.

5 Conclusions

Subcontracting has the potential to help the manufacturing sector industrialize and spread across regions. In India, it has limited takers, and its scope has narrowed down to a few industries and states over the years. Subcontracting in India seems to be a horizontal relationship among the firms. Its primary motive from the principal firms is to maintain flexibility by not expanding its size instead of putting out a part of its work. The principal firm predominantly subcontracts the labour and low skill-oriented work. It also provides principal firms with the benefits of lower-cost manufacturing techniques and protects them from maintaining buffer stocks.

The response of the firms towards the subcontracting system of production is not very impressive, and there have been limited takers for this system in India. It is popular only among the vulnerable section within the small firms. Being less prevalent in the technology-oriented industry reflects that it has a limited role in industrialization. The linkage of small firms with the formal sector does not benefit them in capital accumulation or skill development. Organized manufacturing enterprises take advantage of low-cost production by shedding labour-oriented work that does not require capital or skill.

Therefore, it has remained a tool of survival for the unorganized manufacturing enterprises, especially OAEs, rather than a tool of industrialization. It acts as a survival strategy for resource-scarce or vulnerable firms. The existence of commercial subcontracting is more evident than industrial subcontracting. General pictures that erupt from the above analysis are that the modern dynamic and resource-sufficient firms do not prefer to subcontract. Some forces compel certain firms to adopt the subcontracting mode of production, but that does not necessarily lead to the growth of firms' output and profit share. Firms are not able to generate surplus income and therefore are not able to accumulate capital. It has been a prime obstacle in the

growth of subcontracting firms and MSMEs in general. The role of subcontracting in technology transfer, skill development and productivity growth needs further investigation.

The subcontracting model existing in India does not reflect a sound and sustainable model of production. However, it ensures the survival of poorly endowed and vulnerable firms. Credit provisions through government institutions or the principal firms may help these firms grow their capital and output. Establishing subcontracting exchanges may help larger firms find their ancillaries, establish long-term formal contracts and generate agglomeration effects. The subcontracting leading to higher production scale will help the economy industrialize, and firms growing with higher profit share, benefits society at large.

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Data Availability Data is openly accessible from MoSPI (<http://mospi.nic.in/>).

Declarations

Conflict of interest We hereby declare that we have no conflict of interest to disclose.

Ethics Approval We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

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